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International Academic Research Competition 2026

Research Title : Environmental and Behavioural Predictors of *Aedes* Mosquito Breeding: Development of a Low-Cost Microhabitat-Level Breeding Risk Model in Rangpur, Bangladesh.

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Title: Environmental and Behavioural Predictors of *Aedes* Mosquito Breeding: Development of a Low-Cost Microhabitat-Level Breeding Risk Model in Rangpur, Bangladesh.

1. Abstract

Dengue remains an important public health concern in Bangladesh, but the conditions supporting *Aedes* breeding can vary between urban settings. While major dengue-affected cities such as Dhaka have received considerable research attention, there is limited field-based evidence from Rangpur linking local environmental conditions and human practices with actual *Aedes* breeding [1]–[4]. This study will investigate potential breeding habitats across selected urban settings in Rangpur and identify factors associated with breeding positivity. A stratified multistage approach will cover residential, commercial, healthcare-associated, drainage/waterlogging-prone, canal-associated and urban-peripheral areas. Individual potential breeding habitats will be the primary sampling units. Environmental characteristics and relevant site-management practices will be recorded through field observation and structured questionnaires. Breeding positivity will be defined by the presence of *Aedes* larvae and/or pupae. Descriptive analysis and logistic regression will be used to examine associated factors, followed by exploration of a simple model for estimating relative breeding risk at the microhabitat level. The proposed model will focus on breeding-site risk rather than predicting dengue infection or human transmission.

2. Background and Research Gap

Dengue is an important public health problem in Bangladesh, with substantial variation in reported burden across locations and over time [1], [2], [4]. Dhaka and other major urban areas have received considerable research attention, but comparatively little field-based work has examined the conditions supporting *Aedes* breeding in Rangpur.

Previous studies in Bangladesh have identified several factors associated with *Aedes* breeding, including container characteristics, shade, vegetation, discarded materials and other water-holding habitats [5], [6]. Urbanisation may also increase the availability and distribution of suitable mosquito habitats [7]. However, evidence from Dhaka and other settings may not fully represent the environmental conditions and everyday practices found in Rangpur.

A recent Rangpur study examined residents' knowledge, attitudes and preventive practices regarding dengue [3]. Although useful for understanding community-level behaviour, it did not directly determine whether those practices and local environmental conditions were associated with actual *Aedes* breeding in individual habitats.

The resulting gap is therefore specific: Rangpur has local evidence on dengue-related knowledge and Bangladesh has evidence on breeding-site determinants, but there is limited local evidence connecting environmental conditions and relevant human practices with breeding positivity at the level of individual habitats. This study will address that gap through direct inspection of potential breeding sites and systematic recording of their environmental and management characteristics.

3. Research Questions :

Central Research Question

What environmental and behavioural factors are associated with *Aedes* breeding positivity in individual urban microhabitats in Rangpur, and can these factors be combined into a simple model for estimating relative breeding risk?

Supporting Questions

1. Which types of potential urban microhabitats in Rangpur show *Aedes* breeding positivity?
2. Which environmental characteristics are associated with breeding positivity?
3. Are selected human practices related to the presence or absence of *Aedes* immature stages?
4. Does breeding positivity vary across different urban settings?
5. How well can the identified environmental and behavioural factors distinguish breeding-positive from breeding-negative habitats?

4. Objectives

General Objective

To investigate environmental and behavioural factors associated with *Aedes* breeding in urban microhabitats of Rangpur and explore a low-cost approach for estimating relative breeding risk.

Specific Objectives

1. To identify common urban microhabitats where *Aedes* larvae and/or pupae are present.
2. To assess environmental characteristics associated with breeding positivity.
3. To examine selected practices related to water storage, cleaning and management of potential breeding sites.
4. To compare breeding positivity across different urban settings.
5. To develop and internally assess a simple model for estimating relative microhabitat-level breeding risk.

5. Methodology

10 .1 Study Design, Setting and Sampling

A cross-sectional field study will be conducted in selected urban and urban-peripheral areas of Rangpur. A stratified multistage sampling approach will cover six broad settings: residential, commercial, healthcare-associated, drainage/waterlogging-prone, canal-associated areas including sites around the Shyamasundari Canal, and urban-peripheral/fringe areas.

The individual potential breeding habitat will be the primary sampling unit, allowing environmental and management characteristics to be linked directly with the presence or absence of *Aedes* immature stages. Potential habitats may include water-storage containers, flower pots and trays, rooftop containers, discarded plastic containers, used tyres, construction-related water-holding sites, stagnant drainage water, rainwater accumulation and other suitable habitats.

A pilot survey will be conducted before the main survey to assess field feasibility, identify commonly encountered habitat types and refine sample allocation and field procedures.

5.2 Entomological and Environmental Assessment

Each identified potential breeding habitat will be inspected for *Aedes* immature stages. Larvae and pupae will be recorded, and identification will be performed using appropriate field or laboratory procedures where feasible.

Breeding status will be classified as:

- **Positive:** *Aedes* larvae and/or pupae detected.
- **Negative:** No *Aedes* immature stages detected.
- **Uncertain:** Identification cannot be made with sufficient confidence; specimens will be preserved or handled separately where feasible.

Environmental characteristics will include:

- container type and size;
- water depth and persistence;
- container covering;
- shade or sunlight;
- vegetation and organic debris;
- drainage condition and waterlogging;
- waste accumulation; and
- recent environmental conditions, including rainfall where relevant.

5.3 Behavioural Assessment

A structured questionnaire will assess practices that may influence potential breeding habitats, including water storage, cleaning frequency, container covering, management of flower pots and trays, removal of stagnant water, disposal of unused containers, inspection for larvae or eggs, mosquito-control practices and dengue-related awareness.

Behavioural information will be linked to a habitat only when the respondent is directly responsible for managing it. Public drains, canals and other unmanaged locations will be assessed as environmental sites rather than assigned household practices.

5.4 Quality Control and Contextual Information

Field investigators will receive standardized training before data collection. The questionnaire and field-recording tools will be pilot-tested. Each habitat and specimen will receive a unique identification code, and data will be checked regularly for completeness and consistency with secure storage and backup.

Inter-observer agreement will be assessed where feasible.

Human mobility may also be recorded as contextual information, particularly reported travel between Rangpur and other areas. This information will not be interpreted as evidence of local dengue transmission.

6. Study Variables and Conceptual Framework

Outcome

- Microhabitat-level *Aedes* breeding positivity.

Environmental Predictors

- Container type and size
- Water depth and persistence
- Container covering
- Shade/sunlight
- Vegetation and organic matter
- Drainage/waterlogging
- Waste accumulation
- Recent environmental conditions

Behavioural Predictors

- Water-storage practices
- Cleaning practices
- Container covering
- Stagnant-water removal
- Disposal of unused containers
- Mosquito-control practices

Contextual Variables

- Urban setting
- Location characteristics
- Relevant human mobility information

Conceptual Framework

Urban setting → Environmental conditions + Human practices → Habitat suitability → *Aedes* breeding positivity → Relative breeding-risk estimation

The framework will guide the analysis but will not be used to claim causation from cross-sectional observations. Potential interactions, such as rainfall × water persistence, shade × vegetation, container type × covering and behavioural practices × environmental conditions, will be examined only where biologically reasonable and supported by the data.

7. Statistical Analysis and Model Development

Descriptive statistics will summarize habitat characteristics and breeding positivity. Categorical variables will be analysed using Chi-square or Fisher's exact tests as appropriate, while continuous variables will be analysed according to their distributions.

Binary logistic regression will be used to examine associations between potential predictors and breeding positivity, with adjusted estimates reported with 95% confidence intervals. Confounding, multicollinearity and clustering will be assessed. Cluster-robust or multilevel methods will be considered where appropriate.

Predictors will be selected using biological relevance, previous evidence and the study objectives. Missing data, potential confounding, multicollinearity, the number of outcome events and estimate precision will be considered. The number of predictors will be kept appropriate for the available outcome events to reduce overfitting.

The identified predictors will then be explored in a simple probability model for breeding positivity. Model performance will be assessed using appropriate measures of discrimination and calibration, including ROC-AUC where suitable. Internal validation will use bootstrap or another appropriate resampling approach. Because the model will be developed using Rangpur data, external validation will be required before wider application.

8. Expected Outcomes

The study is expected to:

1. identify common urban microhabitats where *Aedes* breeding occurs;
2. identify environmental and behavioural factors associated with breeding positivity;
3. describe differences in breeding positivity across selected urban settings; and
4. provide a simple framework for estimating relative breeding risk that may support future surveillance and vector-control planning.

9. Significance and Innovation

The study will provide field-based evidence on *Aedes* breeding at the microhabitat level in Rangpur. Instead of examining community knowledge or reported preventive practices alone, it will connect relevant practices and observable environmental conditions with direct inspection of potential breeding habitats.

The key innovation is the shift from broad dengue prediction toward **individual breeding-site risk assessment**. The study will examine whether environmental and behavioural information that can be collected using relatively simple field tools can be combined into a practical framework for estimating relative breeding risk at individual habitats.

This approach is intended to remain low-cost and field-oriented. However, the resulting model would require further validation before it could be considered for wider surveillance or vector-control use. Existing dengue prediction research has largely focused on broader outbreak prediction, whereas this study examines risk at a more local breeding-site level [8].

10. Limitations

The cross-sectional design will limit causal interpretation. Breeding patterns may vary with season and weather conditions [9]. Questionnaire responses may be affected by recall or reporting bias. Some habitats may be inaccessible, and species identification may be uncertain in some specimens. Findings may not be generalizable to all areas of Bangladesh.

Habitats within the same locations may not be statistically independent, which may affect estimates if clustering is ignored. Model performance may also be limited by sample size or the number of breeding-positive habitats. External validation will therefore be necessary before wider application.

11. Ethics

Ethical approval and relevant institutional or local permissions will be obtained before data collection. Questionnaire participation will be voluntary, with informed consent obtained from participants. Personal information will be kept confidential.

The study will involve observation and collection of mosquito immature stages and will not intentionally expose participants to mosquitoes. Data will be reported honestly, without fabrication, falsification, selective omission or manipulation. Relevant conflicts of interest will be disclosed.

12. Work Plan

Period	Activities
Weeks 1–2	Protocol finalization, permissions, training and pilot survey
Weeks 3–6	Main field sampling and questionnaire data collection
Weeks 7–8	Data cleaning and statistical analysis
Week 9	Model development and internal validation
Week 10	Interpretation and final report preparation

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